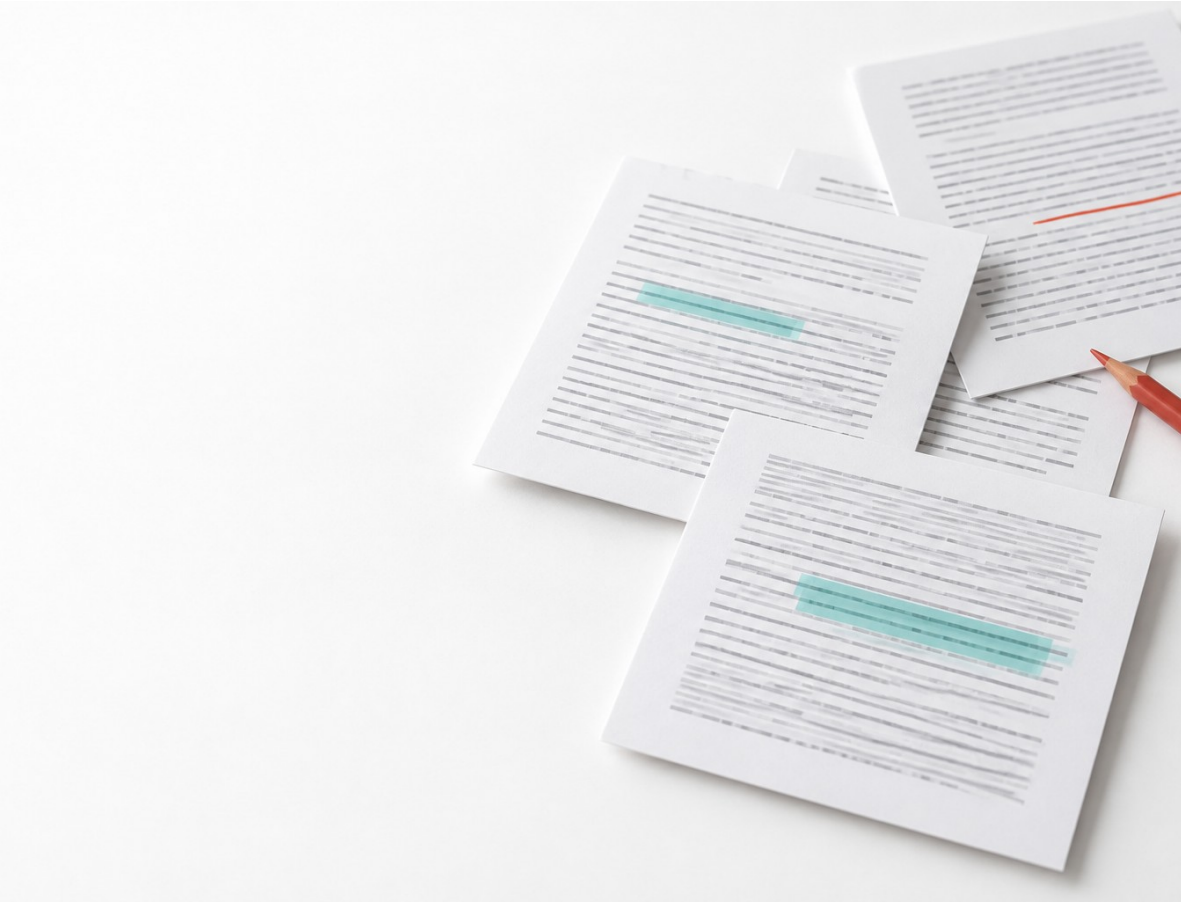
Deep Re-annotation and Controlled Candidate Pools

Re-evaluating ranking models on MS MARCO



One query can have several correct answers

“what causes boiling point to increase”

Passage (shortened)	Original qrels	Human label
7112148 Adding salt raises the boiling point.	Relevant	Relevant
7399382 A non-volatile solute raises the boiling point.	Unjudged	**Relevant**
987600 The boiling temperature of acetone.	Unjudged	Not relevant

qrels = query relevance judgments. Unjudged passages receive zero gain.

STUDY DESIGN

Annotation coverage at two scales

MS MARCO Passage Ranking

8.84 million English passages

1,179

queries with human judgments

33,667 query–passage pairs

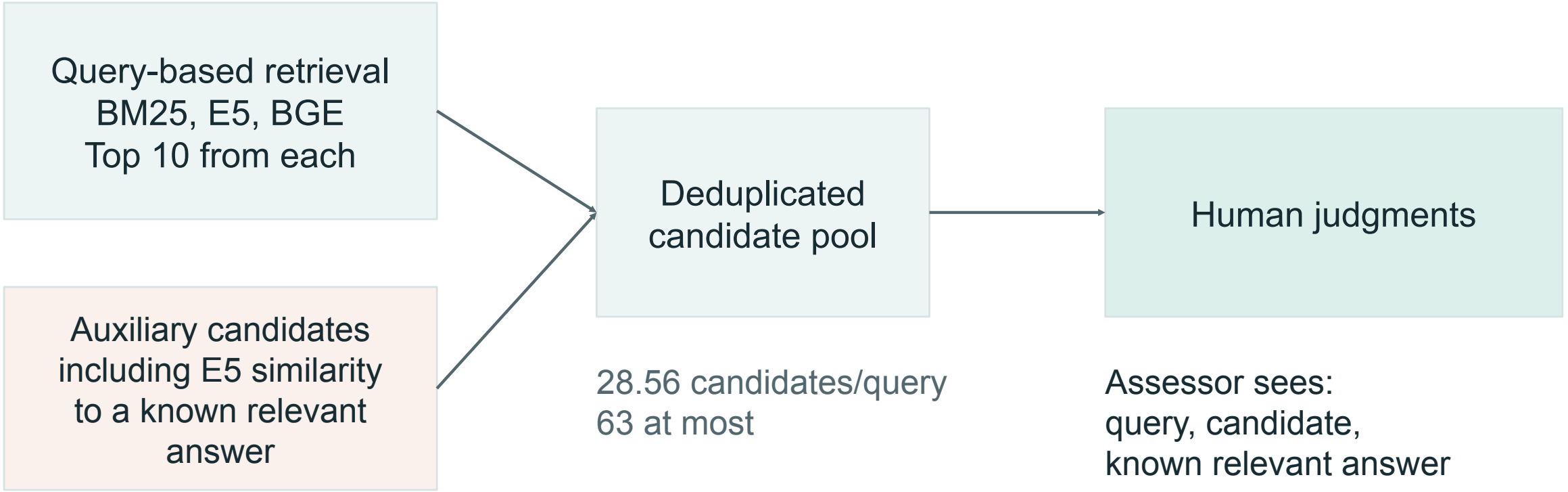
101,093

queries with judge predictions

Candidate depth varies from 10 to 200

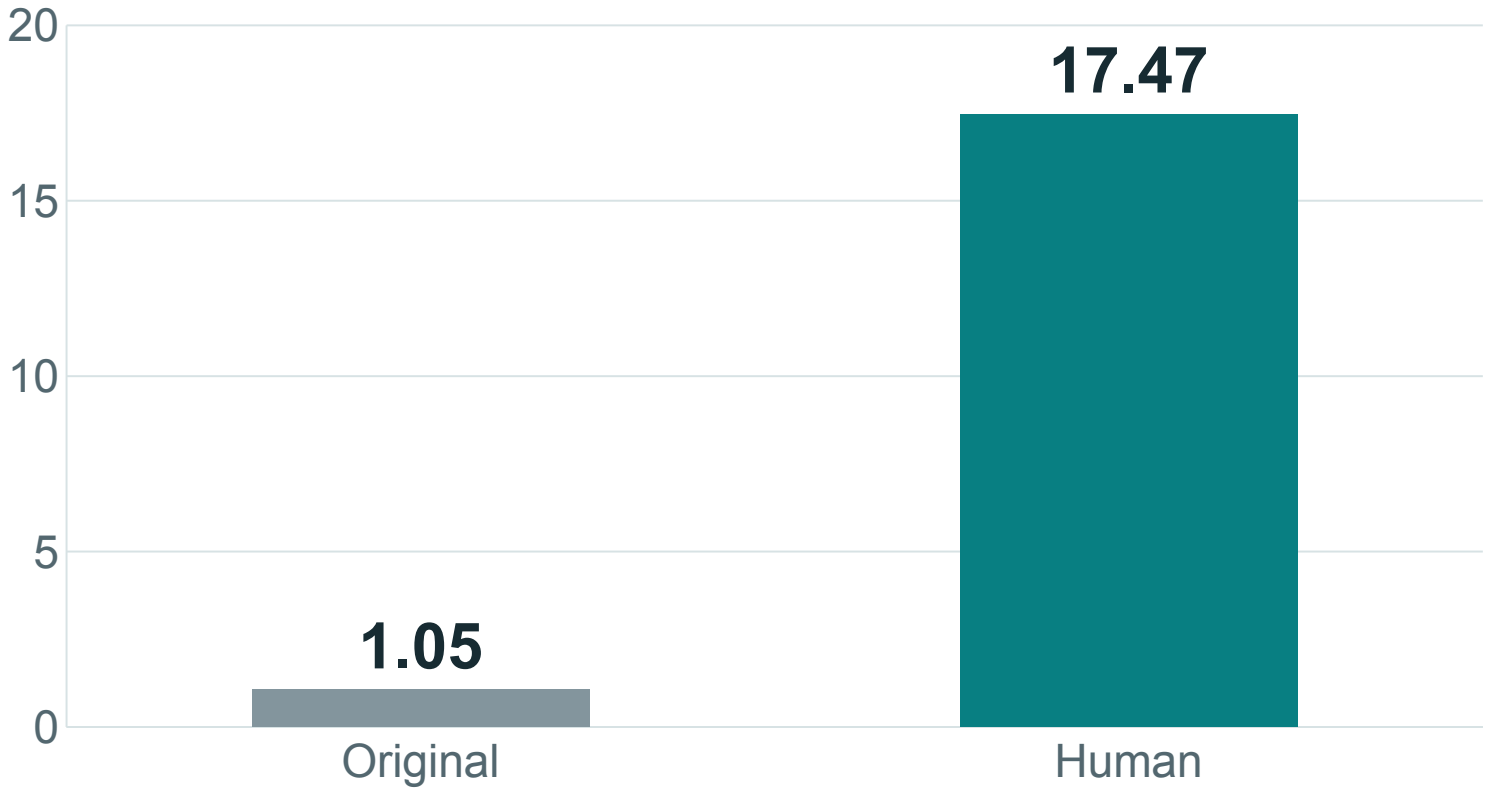
“Deep” refers to judgment coverage within a defined candidate pool.

Pooling and human assessment



BM25: term matching. E5 and BGE: dense bi-encoder retrieval.

Human judgments reveal many more positives



20,597

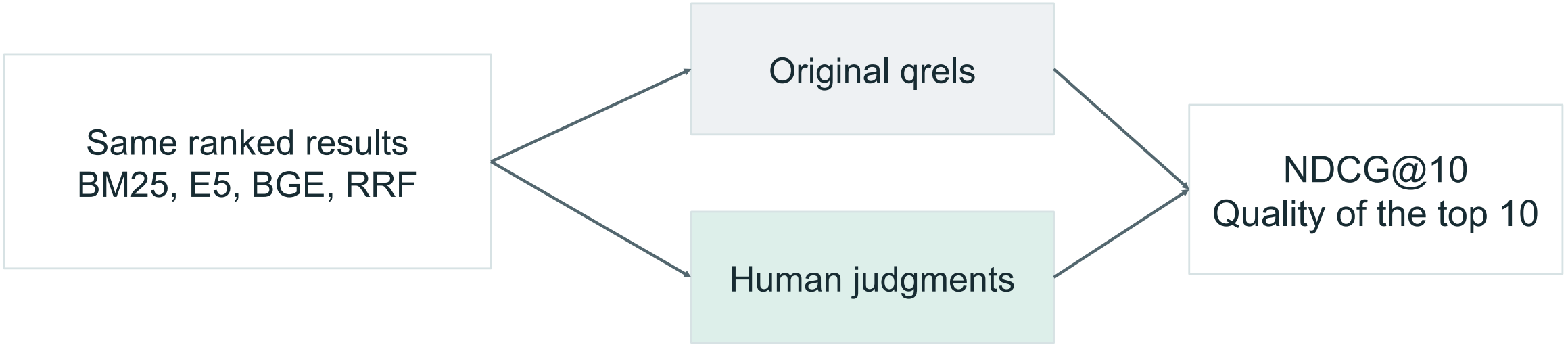
human-positive pairs

1,243 original positives in the same pool

1,179 queries. Counts describe selected candidate pools.

Fixed rankings, different relevance labels

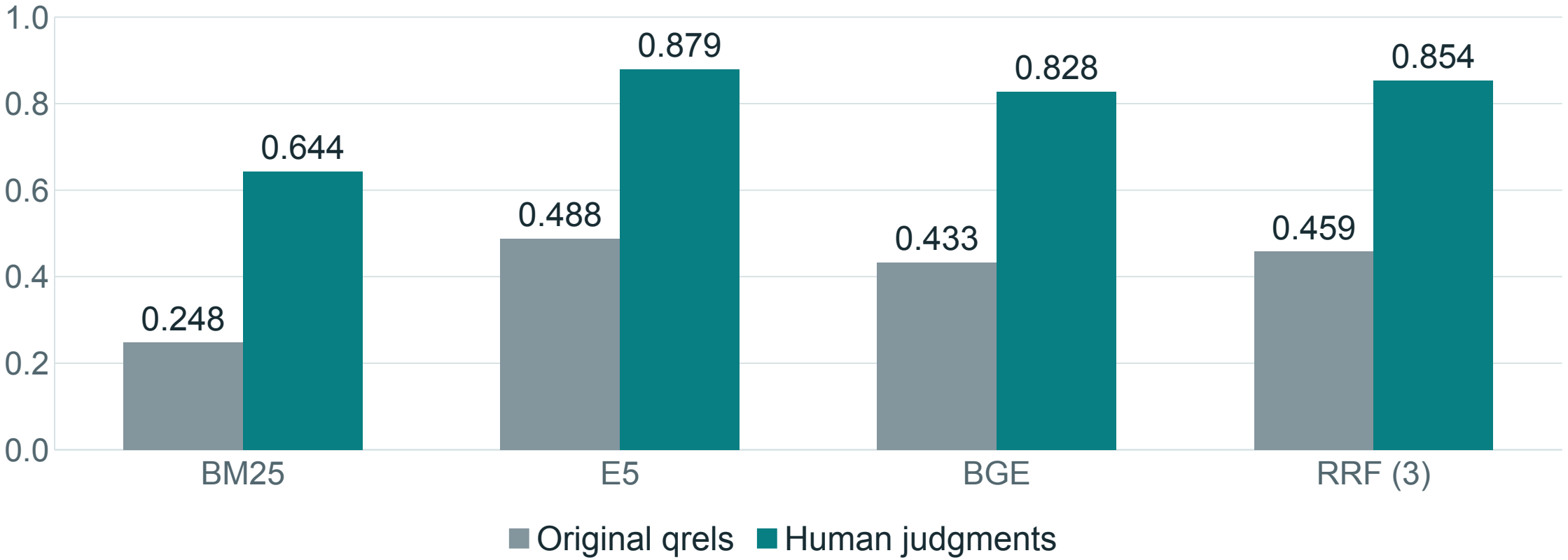
NDCG: normalized discounted cumulative gain



One ideal denominator per query and label set, shared by all systems.

RRF = Reciprocal Rank Fusion. Here: BM25 + E5 + BGE, c = 60.

NDCG rises with denser human judgments



All 1,179 queries. Common pool-based IDCG. Paired changes: $p < 0.001$.

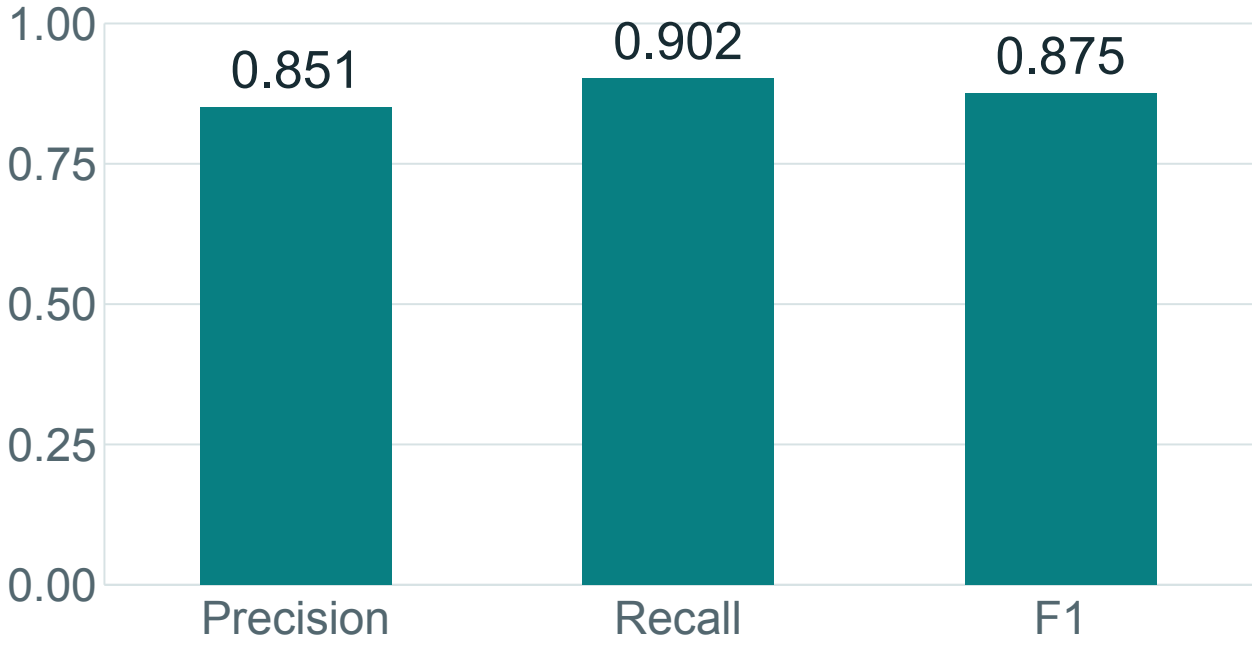
MiniLM cross-encoder as a relevance judge

Input: query text + candidate text

0.907

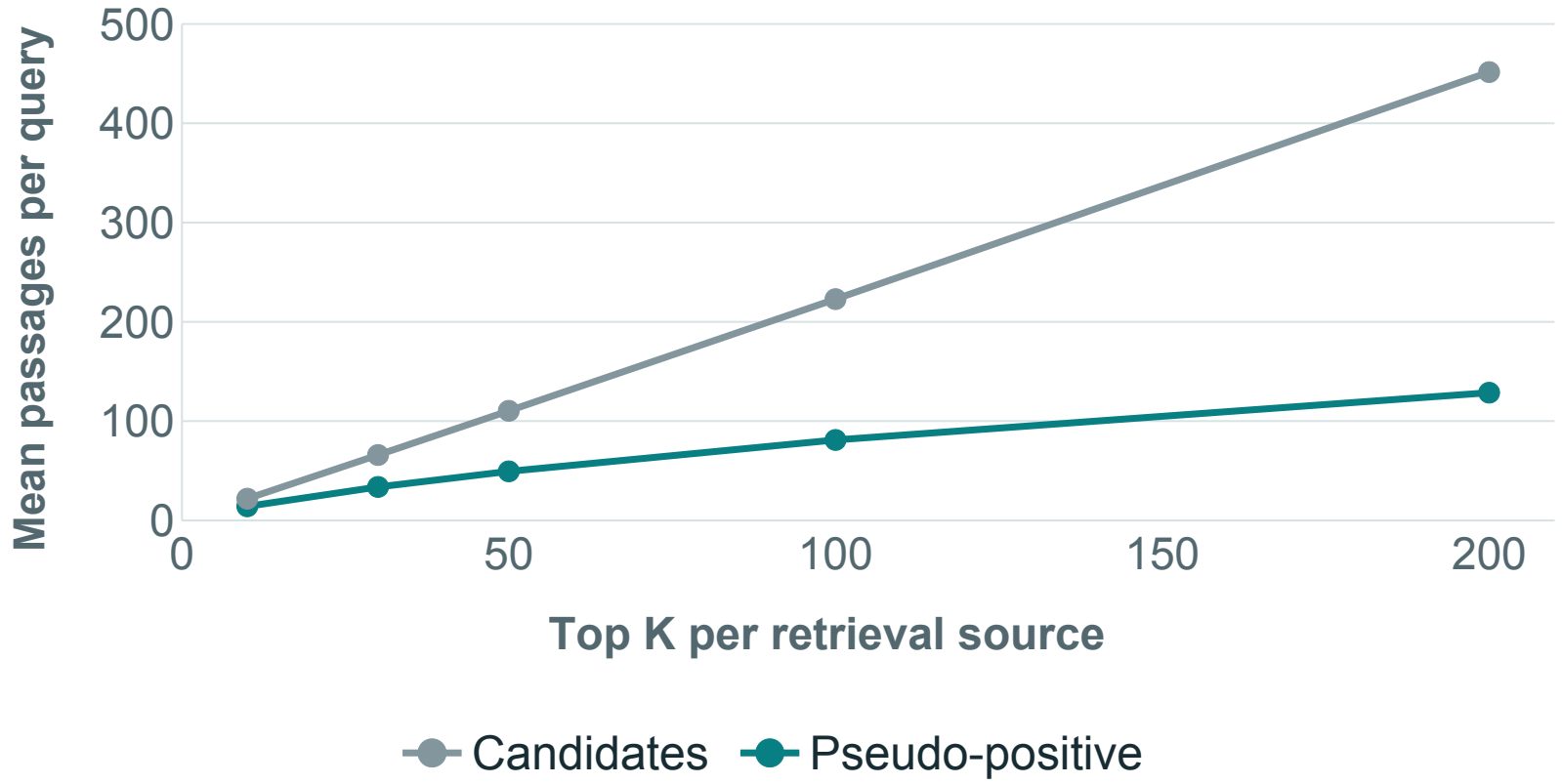
ROC-AUC on held-out queries

177 queries
5,002 query–passage pairs



No reference answer or retrieval scores. Threshold selected on validation.

Pseudo-label counts depend on pool depth

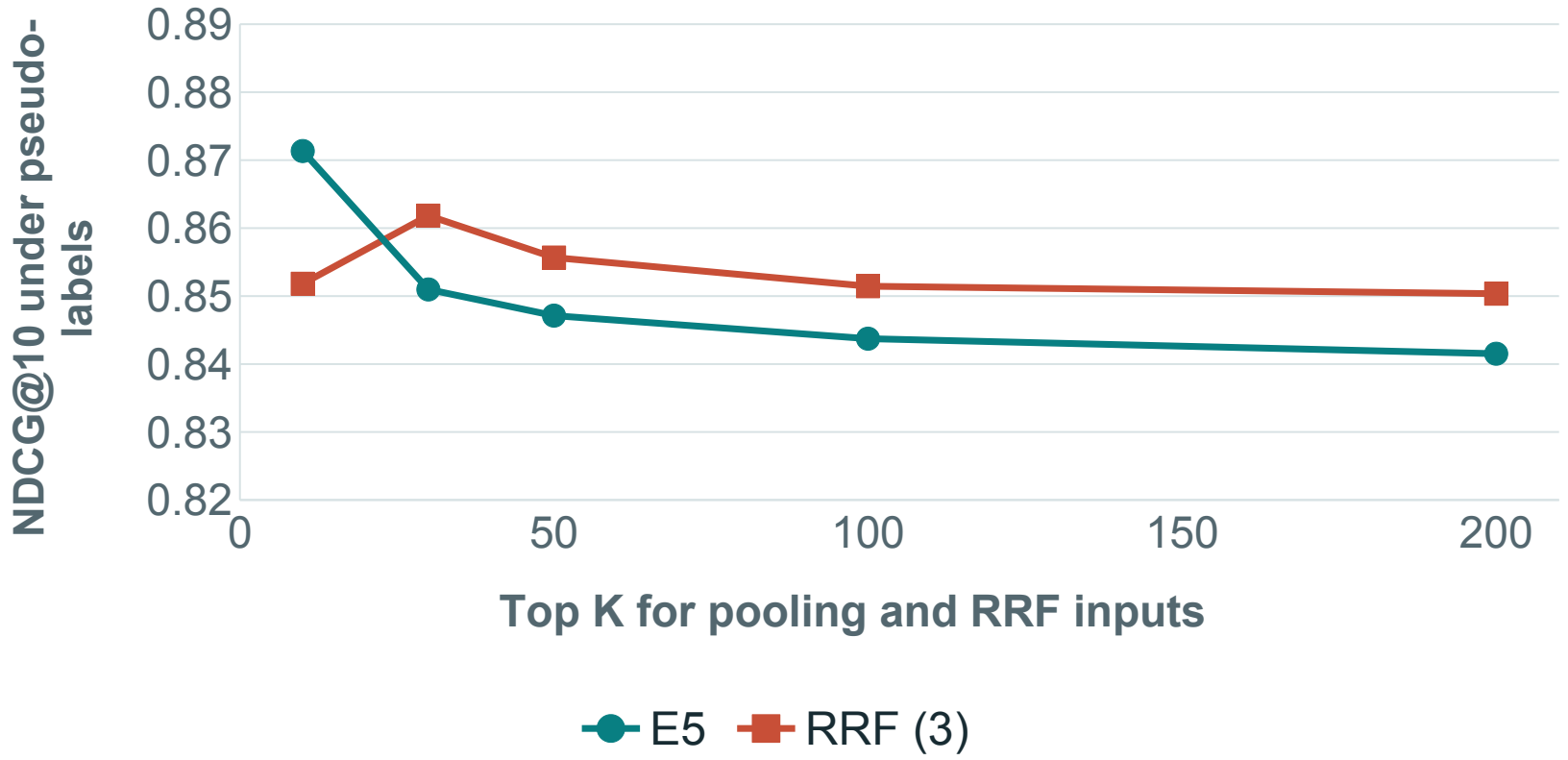


128.6

pseudo-positive passages per query at K = 200

All 101,093 queries. Same judge and threshold at every depth.

RRF depends on the depth of its input lists



RRF minus E5

−0.020

at K = 10

+0.009

at K = 200

Pool depth and RRF input depth both vary. Vertical axis shows a narrow range.

A judge with access to the known answer

Current baseline

Query
Candidate passage

Planned comparison

Query
Known relevant answer
Candidate passage

Stronger cross-encoders and held-out human evaluation

Evidence here is limited to English MS MARCO, selected pools and one judge.

Conclusions

1.05 to 17.47

Human positives per query
inside the annotated pools

Higher NDCG

More complete judgments
for the same ranked results

Conditional RRF gain

A small advantage at deeper inputs
under the judge’s pseudo-labels

NDCG@10 and aggregation rules

NDCG@10 = DCG@10 / IDCG@10

Rule	Definition
Ideal ranking	All positives in the query's candidate pool
No positive judgments	NDCG = 0, query remains in the mean
Unjudged passage	Zero gain, original rank retained
Aggregation	Same query set for every system

Manual uncertainty: 5,000 bootstrap resamples and 10,000 paired sign flips.

APPENDIX B

Judge training and held-out evaluation

Split by query	Queries	Pairs
Train	825	23,543
Validation	177	5,122
Test	**177**	**5,002**

MiniLM-L6 cross-encoder, 2 epochs
Validation-selected raw-score threshold: −0.2187

Held-out pool reranking: NDCG@10 = 0.512 (original), 0.938 (human).

Sensitivity to query selection

Human NDCG@10	All 1,179	Clean 1,156
BM25	0.644	0.644
E5	0.879	0.879
BGE	0.828	0.828
RRF (3)	0.854	0.855

After excluding all manual queries: 99,914 queries

RRF minus E5: -0.020 at K = 10, +0.009 at K = 200

Pseudo-label dependence remains after this exclusion.

Selected references

MS MARCO	Bajaj et al., 2016. arXiv:1611.09268
BM25	Robertson and Zaragoza, 2009. BM25 and Beyond
E5 and BGE	Wang et al., 2022. E5 BAAI/bge-small-en-v1.5 model card
Rank fusion	Cormack, Clarke and Buttcher, 2009. SIGIR
Cross-encoder judge	Nogueira and Cho, 2019. Passage Re-ranking with BERT cross-encoder/ms-marco-MiniLM-L6-v2